Docker 应用部署-RocketMQ

1 安装前准备

1.1 准备安装目录

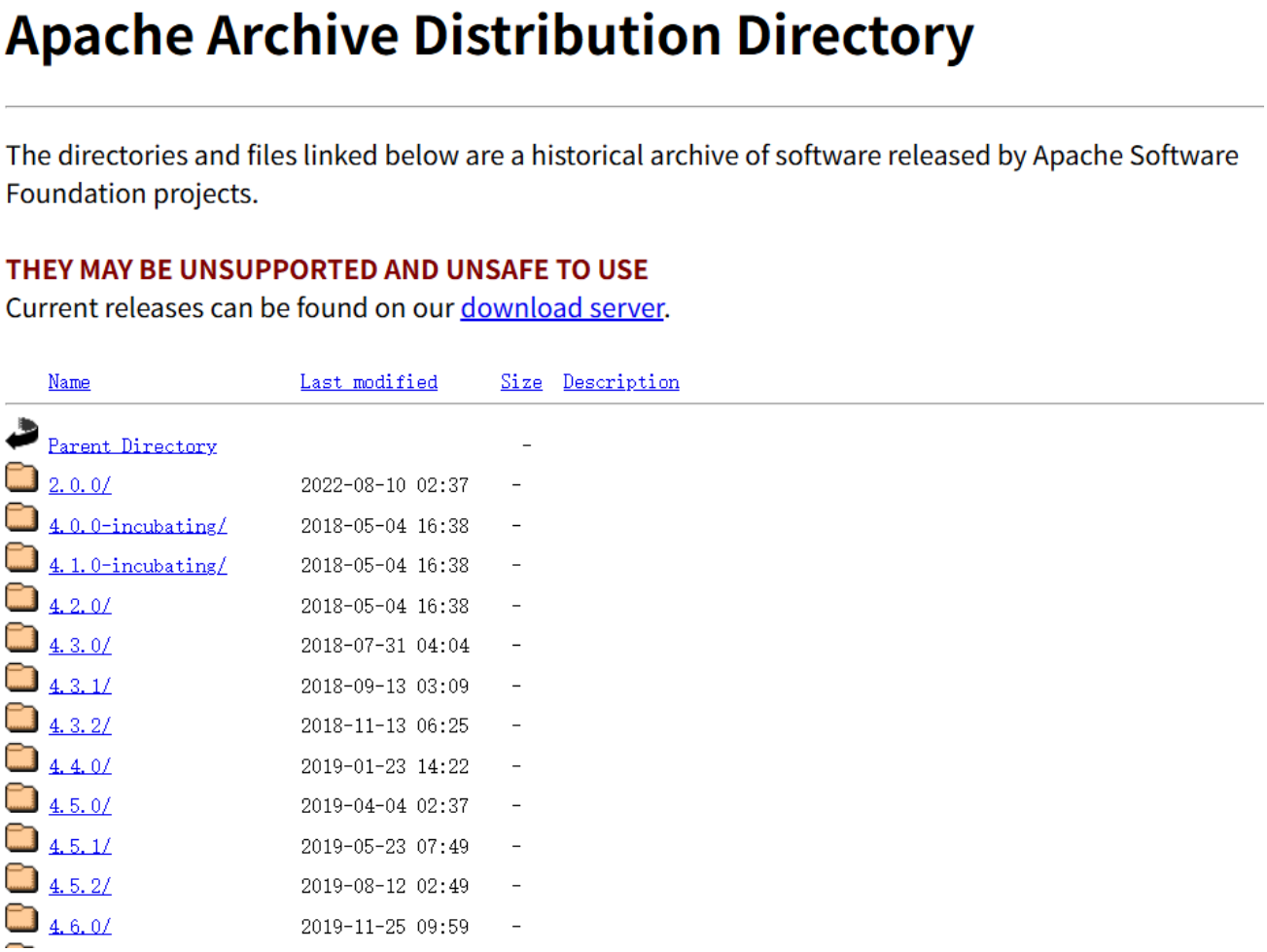
在Linux下面执行下面的命令

```
mkdir /home/rocketmq
cd /home/rocketmq
```

1.2 下载rocketmq

提示：本教程使用的文件我放在资源目录下面了，如果你也是安装5.3.1版本可以直接上传到你的Linux服务器上面，跳过通过官网的下载操作。

在 Rocketmq版本发布页 找到你要下载的版本



Apache Archive Distribution Directory

The directories and files linked below are a historical archive of software released by Apache Software Foundation projects.

THEY MAY BE UNSUPPORTED AND UNSAFE TO USE
Current releases can be found on our download server.

Name	Last modified	Size	Description
Parent Directory		-	
2.0.0/	2022-08-10 02:37	-	
4.0.0-incubating/	2018-05-04 16:38	-	
4.1.0-incubating/	2018-05-04 16:38	-	
4.2.0/	2018-05-04 16:38	-	
4.3.0/	2018-07-31 04:04	-	
4.3.1/	2018-09-13 03:09	-	
4.3.2/	2018-11-13 06:25	-	
4.4.0/	2019-01-23 14:22	-	
4.5.0/	2019-04-04 02:37	-	
4.5.1/	2019-05-23 07:49	-	
4.5.2/	2019-08-12 02:49	-	
4.6.0/	2019-11-25 09:59	-	

点击对应版本进入下载页面，本教程以**5.3.1**为例

Apache Archive Distribution Directory

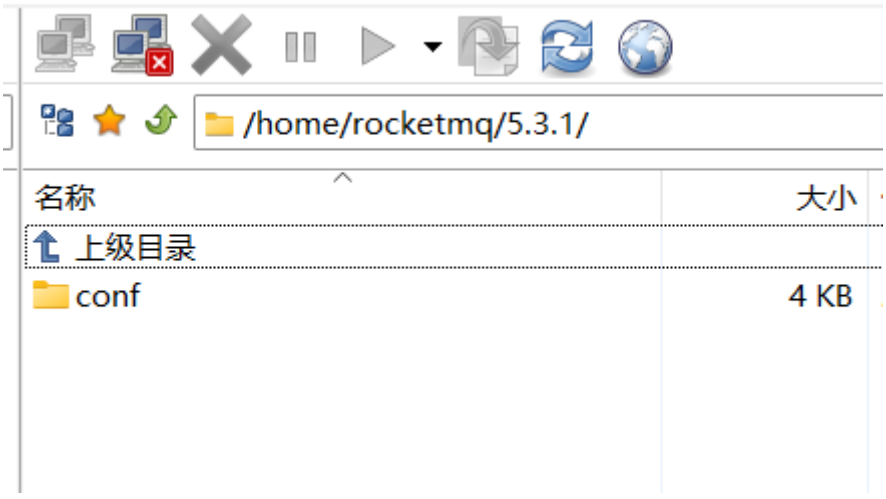
The directories and files linked below are a historical archive of software released b
Foundation projects.

THEY MAY BE UNSUPPORTED AND UNSAFE TO USE
Current releases can be found on our download server.

Name	Last modified	Size	Description
Parent Directory		-	
rocketmq-all-5.3.1-bin-release.zip	2024-10-09 07:32	86M	
rocketmq-all-5.3.1-bin-release.zip.asc	2024-10-09 07:32	836	
rocketmq-all-5.3.1-bin-release.zip.sha512	2024-10-09 07:32	288	
rocketmq-all-5.3.1-source-release.zip	2024-10-09 07:32	9.5M	
rocketmq-all-5.3.1-source-release.zip.asc	2024-10-09 07:32	836	
rocketmq-all-5.3.1-source-release.zip.sha512	2024-10-09 07:32	300	

下载 rocketmq-all-5.3.1-bin-release.zip 文件，下载完成解压文件，然后将conf目录上传到你的Linux服务器上面。

上传目录到`/home/rocketmq/5.3.1`这个目录，上传后目录效果如下图



1.3 准备Broker服务配置

```
# 进入到版本目录
cd 5.3.1
# 创建broker目录
mkdir broker
# 复制配置到broker
cp -r conf/ broker/conf/
```

```
[root@localhost rocketmq]# cd 5.3.1/
[root@localhost 5.3.1]# mkdir broker
[root@localhost 5.3.1]# cp -r conf/ broker/conf/
[root@localhost 5.3.1]# ls broker/conf/
2m-2s-async  2m-noslave  container  dledger         rmq.broke
2m-2s-sync   broker.conf  controller plain_acl.yml  rmq.clien
```

修改`broker/conf`中的`broker.conf`文件，修改参考如下：

```
# 所属集群名称，如果节点较多可以配置多个
brokerClusterName = DefaultCluster
# broker名称，master和slave使用相同的名称，表明他们的主从关系
brokerName = broker-a
# 0表示Master，大于0表示不同的slave
brokerId = 0
# 表示几点做消息删除动作，默认是凌晨4点
deleteWhen = 04
# 在磁盘上保留消息的时长，单位是小时
fileReservedTime = 48
# 有三个值：SYNC_MASTER，ASYNC_MASTER，SLAVE；同步和异步表示Master和Slave之间同步数据的机制
brokerRole = ASYNC_MASTER
# 刷盘策略，取值为：ASYNC_FLUSH，SYNC_FLUSH表示同步刷盘和异步刷盘
# SYNC_FLUSH消息写入磁盘后才返回成功状态，ASYNC_FLUSH不需要
flushDiskType = ASYNC_FLUSH

# 设置broker节点所在服务器的ip地址（非常重要）
# 主从模式下，从节点会根据主节点的brokerIP2来同步数据，如果不配置，主从无法同步，
# brokerIP1设置为自己外网能访问的ip，服务器双网卡情况下必须配置，
# 比如阿里云这种，主节点需要配置ip1和ip2，从节点只需要配置ip1即可
brokerIP1 = 192.168.220.128
# nameServer地址，分号分割
namesrvAddr = 192.168.220.128:9876
# Broker对外服务的监听端口
listenPort = 10911
# 是否允许Broker自动创建Topic，建议线下开启，线上关闭
autoCreateTopicEnable = true
# 是否允许Broker自动创建订阅组，建议线下开启，线上关闭
autoCreateSubscriptionGroup = true
# linux开启epoll
useEpollNativeSelector = true
# 更多配置说明可参考
# https://blog.csdn.net/guyue35/article/details/105439325
```

```
[root@localhost 5.3.1]# vi broker/conf/broker.conf
[root@localhost 5.3.1]# cat broker/conf/broker.conf
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements.  See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
# the License.  You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
#  Unless required by applicable law or agreed to in writing, software
#  distributed under the License is distributed on an "AS IS" BASIS,
#  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
#  See the License for the specific language governing permissions and
#  limitations under the License.

brokerClusterName = DefaultCluster
brokerName = broker-a
brokerId = 0
deleteWhen = 04
fileReservedTime = 48
brokerRole = ASYNC_MASTER
flushDiskType = ASYNC_FLUSH
brokerIP1 = 192.168.220.128
namesrvAddr = 192.168.220.128:9876
listenPort = 10911
autoCreateTopicEnable = true
autoCreateSubscriptionGroup = true
useEpollNativeSelector = true
[root@localhost 5.3.1]#
```

要换成你的ip

1.4 修改目录权限

修改权限前，先创建服务器需要的目录

```
mkdir -p broker/store broker/logs
mkdir -p namesrv/store namesrv/logs
```

容器中的用户ID和组ID分别是 3000 和 3000，如果不给宿主机目录授权限那么容器将会运行不成功的

```
chown -R 3000:3000 ./
```

```
[root@localhost 5.3.1]# mkdir -p broker/store broker/logs
[root@localhost 5.3.1]# mkdir -p namesrv/store namesrv/logs
[root@localhost 5.3.1]# chown -R 3000:3000 ./
[root@localhost 5.3.1]# ll
总用量 4
drwxr-xr-x 5 3000 3000   43 4月   22 16:50 broker
drwxr-xr-x 8 3000 3000 4096 4月   22 16:29 conf
drwxr-xr-x 4 3000 3000   31 4月   22 16:50 namesrv
[root@localhost 5.3.1]#
```

1.5 了解将要安装服务

Name Server：

名称服务充当路由消息的提供者。生产者或消费者能够通过名字服务查找各主题相应的Broker IP列表。

Broker Server：

代理服务器：消息中转角色，负责存储消息、转发消息。

代理服务器在RocketMQ系统中负责接收从生产者发送来的消息并存储、同时为消费者的拉取请求作准备。

代理服务器也存储消息相关的元数据，包括消费者组、消费进度偏移和主题和队列消息等。

Broker 启动时，（默认）实际上会监听3个端口：10909、10911、10912，这三个端口由Broker内部不同的组件使用，作用分别如下：

- **remotingServer：**监听 listenPort 配置项指定的监听端口，默认为 10911
- **fastRemotingServer：**监听端口值 listenPort - 2，即默认为 10909
- **HAService：**监听端口为值为 listenPort + 1，即默认为 10912，该端口用于 Broker 的主从同步

Console Server：

方便以图形化的方式运维MQ服务器情况，同时可以通过控制台对MQ服务器执行一些操作，如：创建主题、发布消息。

2 配置防火墙

注意：如果是云服务器端口放行需要到控制台页面去开放，一般情况操作的系统的防火墙是关闭的，不需要通过下面的命令行开放防火墙。

```
#开放端口
firewall-cmd --add-port 9876/tcp --add-port 10911/tcp --add-port 8081/tcp --add-port 8280/tcp --permanent
#重新加载防火墙
firewall-cmd --reload
```

3 服务编排

在rocketmq目录下面创建`docker-compose.yml`文件，并写入下面内容。

```
services:
  rocketmq-namesrv:
    image: apache/rocketmq:5.3.1
    networks:
      - mq
    container_name: rocketmq-namesrv
```

```
    ports:
      - "9876:9876"
    environment:
      - JAVA_OPT_EXT=-server -Xms128m -Xmx128m -Xmn64m
      - TZ=Asia/Shanghai
    volumes:
      - ./namesrv/logs:/home/rocketmq/logs
      - ./namesrv/store:/home/rocketmq/store
      - /etc/localtime:/etc/localtime:ro
    command: sh mqnamesrv
  rocketmq-broker:
    image: apache/rocketmq:5.3.1
    networks:
      - mq
    container_name: rocketmq-broker
    ports:
      - "10909:10909"
      - "10911:10911"
      - "10912:10912"
      - "8081:8081"
      - "8082:8080"
    environment:
      - JAVA_OPT_EXT=-server -Xms128m -Xmx128m -Xmn64m
      - TZ=Asia/Shanghai
    volumes:
      - ./broker/logs:/home/rocketmq/logs
      - ./broker/store:/home/rocketmq/store
      - ./broker/conf:/home/rocketmq/conf
      - /etc/localtime:/etc/localtime:ro
    command: sh mqbroker -c /home/rocketmq/conf/broker.conf -n rocketmq-
namesrv:9876 --enable-proxy
    depends_on:
      - rocketmq-namesrv
  rocketmq-console:
    image: apacherocketmq/rocketmq-dashboard:2.1.0
    networks:
      - mq
    container_name: rocketmq-console
    ports:
      - "8280:8082"
    environment:
      - JAVA_OPTS=-Drocketmq.namesrv.addr=rocketmq-namesrv:9876 -
Dlogging.level.root=info -Dcom.rocketmq.sendMessageWithVIPChannel=false
      - TZ=Asia/Shanghai
    volumes:
      - /etc/localtime:/etc/localtime:ro
    depends_on:
      - rocketmq-namesrv
      - rocketmq-broker
networks:
  mq:
    name: rocketmq_network
    driver: bridge
```

4 启动与测试

前台启动服务

```
docker compose up
```

首次启动需要下载镜像文件，耐心等待即可

```
[root@localhost 5.3.1]# ls
broker  conf  docker-compose.yml  namesrv
[root@localhost 5.3.1]# docker compose up
[+] Running 0/20
 ⠸ rocketmq-console Pulling
   ⠿ 1e02d32990ad Waiting
   ⠿ 118fd96ad259 Waiting
   ⠿ dbf0c63b062f Waiting
   ⠿ 2bf803abe8d3 Waiting
   ⠿ cf9a91fbcf2e Waiting
   ⠿ 462b7ee4665c Waiting
   ⠿ 843566e9147c Waiting
 ⠸ rocketmq-namesrv Pulling
   ⠸ de44b265507a Downloading [====>                                              ]  2.482MB/29.75MB
   ⠸ ae2feb52a586 Downloading [=============>                                     ]  4.497MB/16.97MB
   ⠸ a833422b45d2 Downloading [======================>                            ]  24.31MB/54.71MB
   ⠸ 29ebaf0b532a Waiting
   ⠸ 5c2a67ca5b13 Waiting
   ⠸ d2c1d3874a91 Waiting
   ⠸ d48f835f0d4d Waiting
   ⠸ b73a351faf9e Waiting
   ⠸ 40b6055ed9f1 Waiting
   ⠸ 4f4fb700ef54 Waiting
 ⠸ rocketmq-broker Pulling
```

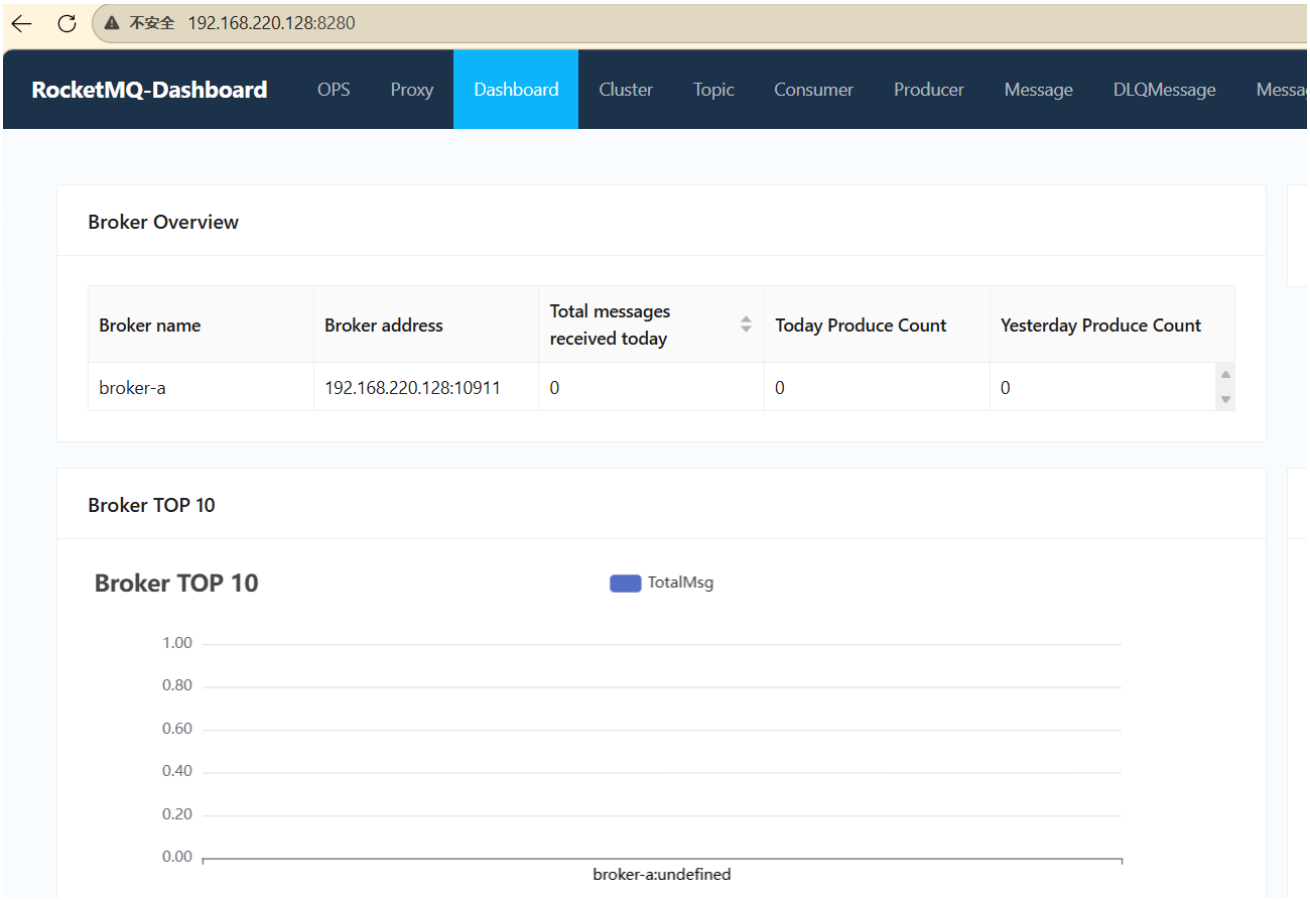
启动成功示意效果如下图

```
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

Boot ::                (v3.4.5)

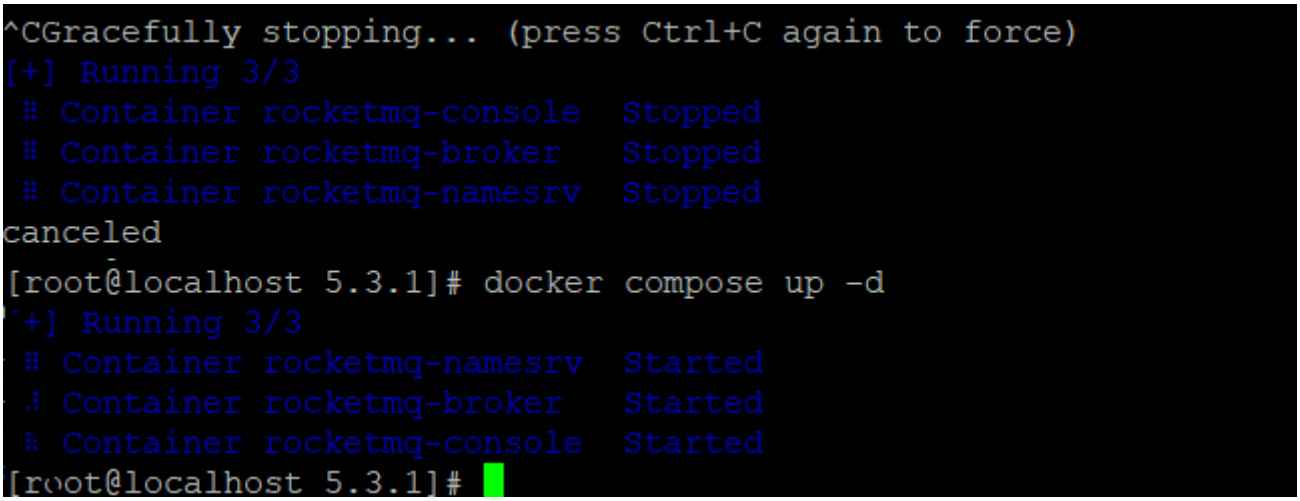
broker-a, 192.168.220.128:10911] boot success. serializeType=JSON and name server is 192.168.220.128:9876
 17:23:26.642] INFO background-preinit - HV000001: Hibernate Validator 8.0.2.Final
 17:23:26.816] INFO main - Starting App v2.1.1-SNAPSHOT using Java 17.0.16 with PID 1 (/rocketmq-dashboard.jar started by root in /)
 17:23:26.818] INFO main - No active profile set, falling back to 1 default profile: "default"
 17:23:30.626] INFO main - Tomcat initialized with port 8082 (http)
 17:23:30.643] INFO main - Initializing ProtocolHandler ["http-nio-8082"]
 17:23:30.649] INFO main - Starting service [Tomcat]
 17:23:30.650] INFO main - Starting Servlet engine: [Apache Tomcat/10.1.40]
 17:23:30.704] INFO main - Initializing Spring embedded WebApplicationContext
 17:23:30.706] INFO main - Root WebApplicationContext: initialization completed in 3659 ms
 17:23:31.397] INFO main - UserMQAdminPoolManager initialized with baseClientConfig for NameServer: rocketmq-namesrv:9876
 17:23:32.337] INFO main - Global AuthenticationManager configured with UserDetailsService bean with name inMemoryUserDetailsManager
 17:23:32.344] INFO pool-3-thread-1 - create MQAdmin instance ClientConfig{namesrvAddr='rocketmq-namesrv:9876', clientIP='172.19.0.4
reads=2, namespace='null', namespaceInitialized=false, namespaceV2='null', accessChannel=LOCAL, pollNameServerInterval=30000, heartb
00, pullTimeDelayMillsWhenException=1000, unitMode=false, unitName='null', decodeReadBody=true, decodeDecompressBody=true, vipChanne
ProxyConfig='{}', mqClientApiTimeout=3000, detectTimeout=200, detectInterval=2000, language=JAVA, enableStreamRequestType=false, sen
artbeatChannelEventListener=true, enableTrace=false, traceTopic='null'} success.
 17:23:32.600] INFO main - Adding welcome page: class path resource [public/index.html]
 17:23:33.477] INFO main - Exposing 14 endpoints beneath base path '/actuator'
 17:23:34.452] INFO main - Starting ProtocolHandler ["http-nio-8082"]
 17:23:34.496] INFO main - Tomcat started on port 8082 (http) with context path '/'
 17:23:34.528] INFO main - Started App in 9.243 seconds (process running for 11.47)
```

连接控制台访问测试，访问地址http://ip:8280/，在集群查看是否可以看到Broker服务，如下图所示。

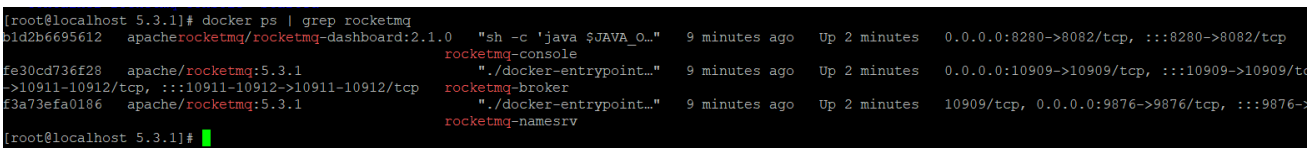


到此可以看到一切正常，下一步可以切换到后台启动了,终端执行`Ctrl + c`结束前台启动，使用下面命令进行后台启动

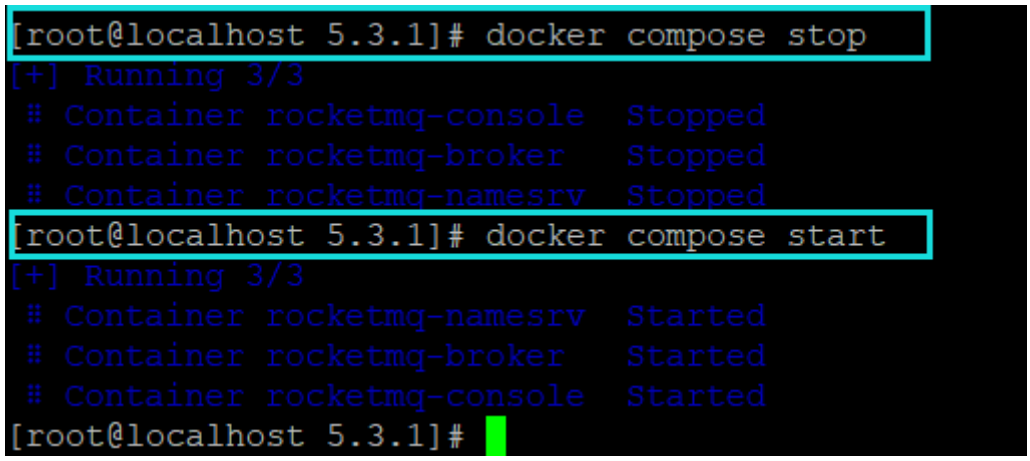
```
docker compose up -d
```



使用docker ps指令查看后台启动情况



因为容器已经创建，后续可以使用stop和start命令来控制容器的停止与启动。


[root@localhost 5.3.1]# docker compose stop
[+] Running 3/3
 # Container rocketmq-console Stopped
 # Container rocketmq-broker   Stopped
 # Container rocketmq-namesrv Stopped
[root@localhost 5.3.1]# docker compose start
[+] Running 3/3
 # Container rocketmq-namesrv Started
 # Container rocketmq-broker   Started
 # Container rocketmq-console Started
[root@localhost 5.3.1]#